



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - Dec 2024

UE19CS352 - Cloud Computing

Total Marks : 100.0

1.a. A pharmaceutical company is working on a highly confidential drug discovery project but also needs to collaborate with external research teams.

What deployment model would suit their needs and justify . (4.0 Marks)

1.b. Which type of cloud users are suited for PaaS? List the advantages and disadvantages of PaaS? (8.0 Marks)

1.c. What are the differences between monolithic and microservices applications? What are the challenges associated with migrating monolithic operations? (8.0 Marks)

2.a. Explain Paravirtualization and Full virtualization with an example
(4.0 Marks)

2.b. What are Shadow Page Tables? How do they work? Explain the advantages and disadvantages of shadow page table.
(8.0 Marks)

2.c. Bring out the comparison between hot migration and cold migration? Explain the different copy techniques of hot migration
(8.0 Marks)

3.a. What is rebalancing? What are the requirements for rebalancing?
(6.0 Marks)

3.b. Discuss 3 desirable properties of the CAP theorem and some of its practical implications.
(6.0 Marks)

3.c. Explain two phase commit protocol with diagram.
(8.0 Marks)

4.a. What steps should a cloud controller take when detecting a failing node in the cluster? (4.0 Marks)

4.b. What is fault tolerance? Name at least two types of failures. What are the considerations for building fault tolerance? (8.0 Marks)

4.c. Explain challenges in multi-leader replication with diagram (8.0 Marks)

5.a. A financial services company is concerned about securing sensitive customer data stored in the cloud. They are worried about unauthorized access and data breaches. How can they enhance security? (4.0 Marks)

5.b. what is a reverse proxy?How it works?

(8.0 Marks)

5.c. What is Cloud bursting? List some common cloud bursting scenarios. Explain how Cloud Bursting can be Beneficial to Cloud Users? (8.0 Marks)